MIL-C-15567(BuOrd)

11 September 1950
SUPERSEDING
Bureau of Ordnance
Specification 4C6(Crd)
7 May 1948

MILITARY SPECIFICATION

CELLULOSE - NITRATE - PLASTIC

(CELLULOID OR PYROXYLIN TYPE)

(FOR USE IN AMMUNITION)

---

1. SCOPE AND CLASSIFICATION

1.1 Scope. - Composition A (stabilized) cellulose-nitrate-plastic is intended for use in powder wads and spacers. Composition B cellulose-nitrate-plastic discs are intended for use in fuzes, tracers, and primers.

1.2 Compositions and Shapes. Cellulose-nitrate-plastic shall be furnished in the following composition and shapes as specified in contract or order (see 6.1):

Composition A (Stabilized)
Sheets
Formed

Composition B
Discs

2. APPLICABLE SPECIFICATIONS

2.1 Specifications.- Navy Department General Specifications for Inspection of Material of the issue in effect on date of invitation for bids, forms a part of this specification.

(Navy.- Copies of Military (including Joint Army-Navy and National Military Establishment Specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D.C., except that activities of the Armed Forces should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying number or symbol should be stipulated when requesting copies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:19Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material. -

3.1.1 Composition A cellulose-nitrate-plastic shall be a homogeneous stabilized nitrocellulose-camphor-urea composition of good quality. Ingredients used shall conform to the following requirements.

3.1.1.1 Nitrocellulose. - The nitrocellulose used shall be prepared from cotton cellulose and shall be thoroughly purified by a treatment of at least 8 hours boiling at 100°C in acid water with an equivalent acid content equal to at least 0.05 percent sulphuric acid. The nitrogen content shall be approximately 11 percent and the solubility in acetone shall be 99 percent. The nitrocellulose shall pass the standard 135°C heat test with normal methyl violet paper of not less than 30 minutes.

3.1.1.2 Camphor. - The camphor shall be of a good commercial grade. A 20 gram sample shall show no discoloration when heated in a glass cup for 30 minutes in an oil bath at 150°C.

3.1.1.3 Urea. - The urea shall be white or light cream in color, neutral or very slightly alkaline, and free from ammoniacal odor. The melting point shall be 132 to 133°C. Ten grams shall dissolve completely in 250 ml of ethyl alcohol with only slight turbidity and the solution on standing at room temperature shall not develop a marked brownish coloration.

3.1.2 Composition B cellulose-nitrate-plastic shall be made from materials of such quality as to produce discs in compliance with the requirements of 3.4.2.

3.2 Color. - Cellulose-nitrate-plastic shall be clear and transparent with a color no darker than a light amber. No coloring matter shall be added.

3.3 Dimensions. - Discs and formed cellulose-nitrate-plastic shall conform to dimensions shown on the drawings furnished with the contract or order. Sheet material shall conform to the dimensions shown in the contract or order.

3.4 Chemical composition. -

3.4.1 Composition A cellulose-nitrate-plastic shall conform to the following compositions by weight:

Nitrocellulose - - - -	approximately 3 parts
Camphor - - - -	approximately 1 part
Urea - - - -	0.75 ± .05 percent of dry weight of nitrocellulose
Solvent - - - -	amount sufficient for processing

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:19Z
Check the source to verify that this is the current version before use.

3.4.2 Composition B cellulose-nitrate-plastic discs shall conform to the following composition:

Nitrocellulose (Nitrogen content - not less than 11 percent) - 72 to 78 Percent
Camphor - Remainder
Mineral ash - 1 percent maximum
Residual solvent (1) - as follows

Disc Thickness	Percent Residual Solvent
Up to 0.060 inch	0.83 - 1.67
0.060 to 0.100 inch	1.67 - 2.50
0.100 to 0.200 inch	2.50 - 3.33
0.200 to 0.300 inch	3.33 - 5.00

(1) Residual solvent, under this specification, is defined as the loss in weight expressed as a percentage of the original sample when heated at 100°C for 3 hours.

3.5 Stability. - Composition A cellulose-nitrate-plastic shall withstand the stability test described in 4.4.1.

3.6 Formability. - Composition A cellulose-nitrate-plastic sheets shall be capable of being formed into cups and retain their shape as shown on the drawings furnished with the contract or order (see 6.1).

3.7 Workmanship. - The workmanship shall be such as to produce a bubble-free material with a smooth surface. Discs shall be flat, free from burrs, nicks, miscuts, wrinkles, buckles or other injurious defects.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot. - For purposes of sampling, a lot shall consist of all material of the same composition, shape and from the same batch submitted for inspection at the same time.

4.2 Sampling. -

4.2.1 Composition A. - Inspection samples for chemical tests shall consist of 8 ounces of cellulose-nitrate-plastic selected by the inspector from each lot. In addition 4 ounces of camphor, 4 ounces of urea, and 8 ounces of nitrocellulose (dry weight) shall be selected by the inspector to represent the raw material used for the production of each lot of finished cellulose nitrate plastic.

4.2.2 Composition B. - Samples for chemical analysis shall be selected from each lot. At least 1 percent of the discs in each box shall be selected for visual inspection.

4.3 Inspection. -

Source: https://assist.dla.mil - Downloaded: 2026-10-18T06:19Z
Check the source to verify that this is the current version before use.

4.3.1 Place of inspection. - Unless otherwise specified, samples of Composition A cellulose-nitrate-plastic shall be forwarded to the Naval Powder Factory, Indian Head, Maryland, for test. Composition B, discs, shall be inspected at the point of delivery unless otherwise specified.

4.3.2 Visual inspection and gaging. - Inspection shall be made on representative shapes, sheets, or discs selected by the inspector from each lot to determine compliance with 3.2, 3.3 and 3.7.

4.4 Tests. -

4.4.1 Stability (Composition A only). - A sample of 2 grams of cellulose-nitrate-plastic cut in quarter inch squares shall be placed in a pyrex test tube (3/4 x 2-3/4 inches) which is covered with a perforated porcelain plate (crucible disk). The plate shall hold in place a strip (5 cm. x 0.8 cm.) of N/10 methyl violet paper projecting into the tube. The tube is fitted into a stainless steel support which is then placed in a quart fruit jar containing 25 grams of K_2SO_4 and 100 ml of distilled water to give a relative humidity of 95 percent. The jar is closed with a stainless steel top and rubber gasket. The jar and contents are placed in a 65.5°C surveillance oven and the time required for the complete decolorization of the paper is recorded. Brown fumes of varying amount will also appear. The time for the color change to a full white paper shall be not less than 30 days.

4.4.2 Chemical tests. - Tests of samples of plastic and of chemical components taken as indicated in 4.2.1 and 4.2.2 shall be made to determine compliance with the requirements of 3.1 and 3.4.

4.4.2.1 Test for residual solvent (Composition B only). - A sample of 10 grams of plastic is cut into small pieces and weighed. The weighed sample is then heated for 3 hours at 100 $\pm$ 1°C, cooled, and then weighed again. The loss in weight is computed to percent residual solvent.

4.4.3 Formability (Composition A only). - Compliance with the requirements of 3.6 shall be determined by immersing the sheet or blank in boiling water and forming to drawing dimensions. The formed piece shall retain its shape after cooling and removal from the forming die.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Cellulose-nitrate-plastic shall be packaged in containers of sufficient strength and construction to prevent damage in shipment and to insure acceptance by common or other carrier for safe transportation at the lowest rate, to the point of delivery. Discs shall be packaged in boxes of not more than 5000.

5.2 Marking of shipping containers. - Unless otherwise specified in the contract or order, shipping containers shall be marked with

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:19Z
Check the source to verify that this is the current version before use.

the name and size of the material, the type, the quantity contained therein, the name of the manufacturer or contractor, the number of the contract or order and the gross weight.

5.3 Labels. - Shipping containers shall be marked and labeled as prescribed by the Interstate Commerce Commission.

6. NOTES

6.1 Ordering information. - Invitation for bids and contracts or orders should specify the following:

Title, number and date of this specification.
Composition and dimensions of sheet or disc (see 1.2 and 3.3).
Drawing number of formed part (see 4.4.3).

6.2 Trade names. - Cellulose-nitrate-plastic (not necessarily complying with the requirements of this specification) are marketed under various trade names, including the following: Celluloid, Pyralin, Viscoloid, Nixonoid, Amarith, Fiberloid and Nitron.

6.3 Superseded specifications. - Composition A (stabilized) cellulose-nitrate-plastic was formerly covered by Ordnance Specification 761, Celluloid (or Pyralin) for Wads and Spacers. Composition B replaces Ordnance Specification 2505, Celluloid Discs.

6.4 Patent notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights, or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:19Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEM PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:19Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER
NAVAL ORDNANCE SYSTEMS COMMAND HEADQUARTERS
ATTN: CODE 9343
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T06:19Z
Check the source to verify that this is the current version before use.